The Coors Effect

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Abstract

Coors Field is one of the most extreme offensive environments in Major League Baseball due to its high altitude and reduced air density, which alter both pitch movement and batted-ball flight. Traditional park factor adjustments account for average run inflation but typically assume that park effects apply uniformly across players, potentially masking differences in how individual skill profiles interact with the environment.

This study examines how both hitter characteristics and pitching profiles influence performance at Coors Field using plate appearance-level Statcast data from the 2015-2025 MLB seasons. Offensive output is measured using run value, which captures the change in expected runs for each plate appearance. Hitter skill is characterized using a combination of batted-ball quality, launch angle tendencies, spray direction, and plate discipline metrics derived from non-Coors data to isolate underlying player ability. Two complementary approaches are used: a clustering-based framework that groups players into discrete archetypes, and a continuous trait-based model that allows individual skills to interact directly with the Coors environment. On the pitching side, pitch shape clusters and pitcher archetypes are constructed using velocity, spin rate, and movement characteristics.

Mixed-effects regression models are used to estimate Coors Field effects while accounting for batter and pitcher identity as well as season-to-season run environment variation. Results show that Coors Field significantly increases offensive production across all hitter groups, with only modest differences across hitter archetypes. Similarly, pitch shape clusters and pitcher archetypes exhibit limited variation in Coors impact, suggesting that altitude does not disproportionately affect specific pitch types or pitching repertoires. In contrast, the continuous hitter trait model shows that Coors performance is driven primarily by top-end contact quality and batted-ball profile characteristics, particularly 90th percentile exit velocity and ground ball rate, and provides a substantially better fit to the data.

To explain these patterns, the study further analyzes pitch-level physics to examine how pitch characteristics change at altitude. Results show that induced vertical break is consistently reduced across nearly all pitch types, while changes in horizontal movement vary by pitch type. These changes affect pitch effectiveness broadly rather than selectively, providing a physical explanation for the uniformity of Coors effects across pitch shapes and pitcher archetypes.

Overall, performance at Coors Field is better explained by continuous skill traits and underlying pitch physics than by discrete player classifications. These findings have implications for player evaluation, roster construction, and the interpretation of park-adjusted metrics.

1. Introduction

Coors Field has long been recognized as the most extreme offensive environment in Major League Baseball. Located over 5,000 feet above sea level in Denver, the stadium's high altitude produces lower air density than most other MLB parks. These conditions reduce aerodynamic drag and weaken the Magnus force on pitched balls, resulting in less pitch movement and increased batted-ball carry. As a result, games at Coors Field consistently produce higher run scoring than league averages.

Because of these differences, analysts commonly apply park factor adjustments to account for the offensive inflation associated with Coors Field. However, traditional park factors assume that these effects apply uniformly across players. In reality, both hitters and pitchers exhibit distinct skill profiles, and the impact of Coors Field may vary based on underlying player characteristics.

For example, hitters with elevated batted-ball profiles may benefit more from increased carry, while reduced pitch movement at altitude may affect pitch effectiveness differently depending on pitch shape and arsenal composition. These considerations raise a broader question: are Coors Field effects driven by discrete player archetypes, specific skill traits, or more fundamental changes in pitch physics at altitude?

This study investigates how both hitter traits and pitching characteristics interact with the Coors Field environment using Statcast data from multiple MLB seasons. Hitter traits are used to construct offensive archetypes and to estimate a continuous trait interaction model that evaluates how specific skills influence performance at Coors Field. On the pitching side, pitch shape clusters and pitcher archetypes are used to assess whether certain pitch types or pitching repertoires are more sensitive to altitude.

To further explain these patterns, the study incorporates a pitch physics analysis that quantifies how pitch movement changes at Coors Field and how these changes relate to run prevention. By integrating hitter traits, pitcher characteristics, and pitch-level physics, this approach provides a more complete understanding of how Coors' extreme environment influences performance.

This study aims to move beyond traditional park factor adjustments by identifying the underlying drivers of Coors Field effects and clarifying whether variation in performance is best explained by player archetypes, continuous skill traits, or the physical properties of the environment itself.

2. Literature Review

Prior research has examined Coors Field primarily by focusing on the environmental mechanisms that contribute to its unusually high offensive production. One line of work studies how atmospheric and meteorological conditions influence batted-ball outcomes in Denver. Chambers, Page, and Zaidins investigate whether baseballs travel as much farther at Coors Field as theoretical models of reduced air resistance predict. Using multi-season fly-ball distance data, they find that while batted balls do travel farther on average, the increase is smaller than simple physics-based models suggest. The authors attribute this gap to additional factors such as wind patterns and ballpark geometry, which can partially offset the distance gains expected from lower air density.

Other research has focused on how altitude affects pitching outcomes. Schaffer and Chandran analyze pitcher performance across ballparks using a statistical framework that groups stadiums by elevation. Their results show that pitching metrics such as opponent on-base percentage and earned run average increase systematically with elevation, with Coors Field representing the most extreme case. These findings suggest that altitude-related effects such as reduced pitch movement and command challenges contribute to the park's offensive environment. However, their analysis evaluates pitching performance using aggregate statistics and does not examine how specific pitch characteristics or pitcher profiles respond to these conditions.

While this body of work clearly establishes that Coors Field produces substantially different offensive outcomes than other ballparks, most prior studies evaluate these effects using park-level or season-level statistics. As a result, relatively little research has examined how the Coors environment interacts with player-specific characteristics at a granular level.

This study builds on existing research by shifting the unit of analysis to plate appearance–level data and incorporating player-specific traits derived from Statcast measurements. In addition to evaluating hitter characteristics such as contact quality and batted-ball tendencies, the analysis also considers pitching-side factors, including pitch shape, pitch usage, and movement profiles. By integrating hitter traits, pitcher characteristics, and pitch-level physics, this study aims to determine whether Coors Field effects are driven by discrete player archetypes, continuous skill traits, or broader changes in pitch behavior at altitude.

3. Data

This study uses pitch-level Statcast data from Major League Baseball covering the 2015-2025 seasons. Statcast provides detailed measurements for each pitch and batted ball, including exit velocity, launch angle, spray direction, pitch characteristics, and game context. After restricting the sample to regular-season games and removing observations with missing data, the dataset contains approximately 1.5 million plate appearances.

Offensive performance is measured using run value, which evaluates each plate appearance based on its impact on expected runs given the base-out state and game context. This approach allows offensive contributions to be measured at the plate appearance level rather than relying on aggregate statistics such as batting average or slugging percentage.

Hitter traits are constructed from aggregated Statcast metrics describing batted-ball characteristics and plate discipline tendencies. To isolate underlying player ability, these traits are calculated using only non-Coors observations. The resulting variables capture multiple dimensions of offensive skill, including contact quality, batted-ball profile, directional tendencies, and swing behavior. Key measures include 90th percentile exit velocity, barrel rate, ground-ball and fly-ball rates, pull tendencies, and plate discipline metrics such as zone swing rate, zone contact rate, strikeout rate, and walk rate.

These traits are aggregated at the player-season level and standardized before being used in clustering procedures to identify offensive archetypes. Plate appearance observations are then linked to these player-level traits and used in subsequent modeling to estimate how different skill profiles perform in the Coors Field environment.

4. Methodology

1. Hitter Archetype Interaction Model

The analysis begins by grouping hitters into offensive archetypes based on underlying skill traits derived from Statcast measurements. Traditional outcome-based statistics provide limited insight into the processes that generate offensive production, while Statcast data allows hitter profiles to be constructed using variables that capture contact quality, batted-ball tendencies, directional behavior, and plate discipline.

The trait set includes measures of batted-ball quality and swing behavior, such as 90th percentile exit velocity, barrel rate, ground-ball and fly-ball rates, pull tendencies, and spray variability, along with plate discipline metrics including zone swing rate, zone contact rate, swinging strike rate, strikeout rate, and walk rate. Sprint speed is also incorporated to capture the contribution of baserunning and infield hit probability. Together, these variables describe multiple dimensions of offensive skill, including how frequently hitters elevate the ball, how hard they hit it, and how selectively they swing.

Traits are calculated at the player-season level using only non-Coors observations and are standardized prior to clustering so that variables measured on different scales do not disproportionately influence the results. To reduce dimensionality and account for correlation among input variables, principal component analysis (PCA) is applied to the standardized trait matrix, and the leading components explaining the majority of variance are retained.

Hitters are then grouped using a Gaussian Mixture Model (GMM) applied to the reduced feature space. This approach models each cluster as a probabilistic distribution, allowing for flexible cluster shapes and accommodating overlap between offensive profiles. The number of clusters is selected based on model fit criteria and interpretability, resulting in a small set of archetypes that capture distinct combinations of power, contact quality, and approach.

To evaluate whether these archetypes experience different outcomes at Coors Field, a mixed-effects regression model is estimated using plate appearance-level run value as the response variable. The model includes an indicator for Coors Field, categorical indicators for hitter archetype, and interaction terms between the two, allowing the estimated park effect to vary across offensive profiles.

The archetype interaction model can be written as:

$$RunValue_i = \beta_0 + \beta_1 * Coors_i + \beta_2 * Cluster_i + \beta_3 * (Coors_i * Cluster_i) + \beta_4 * Rockies_i + u_{batter[i]} + u_{pitcher[i]} + u_{year[i]} + \epsilon_i$$

Random intercepts for batter and pitcher identity are included to account for differences in baseline player ability. These random effects ensure that the estimated park effects are not driven by the presence of particularly strong hitters or weak pitchers in specific environments. Year fixed effects are also included to capture seasonal variation in league run environments. The Rockies indicator is included to separate park effects from team-specific influences.

Model fit is evaluated by comparing this specification to a reduced model that excludes the interaction terms between Coors Field and hitter archetype. A likelihood ratio test is used to assess whether allowing the Coors effect to vary across archetypes significantly improves explanatory power.

2. Hitter Trait Interaction Model

While clustering provides an intuitive way to summarize hitter types, it reduces continuous variation in player traits into a small number of discrete groups. This simplification can obscure more granular relationships between individual skill characteristics and park effects. To address this limitation, the analysis also estimates a continuous trait interaction model that allows specific hitter attributes to directly influence the magnitude of the Coors Field effect.

In this specification, offensive outcomes are modeled as a function of selected hitter traits and their interactions with the Coors Field indicator. Rather than assigning players to a single archetype, this approach evaluates how variation in individual skill dimensions relates to changes in performance at altitude.

The model focuses on key dimensions of offensive production derived from Statcast data, which are 90th percentile exit velocity, ground-ball and fly-ball rates, average exit velocity on airborne contact, pull air rate, and sweet spot rate. These traits are chosen to capture differences in how frequently hitters elevate the ball, how hard they hit it, and the direction and quality of their contact.

The trait interaction model can be written as:

$$RunValue_i = \beta_0 + \beta_1 * Coors_i + \sum_k \beta_k * Trait_{k,i} + \sum_k \gamma k * (Coors_i * Trait_{k,i}) + \beta_4 * Rockies_i + u_{batter[i]} + u_{pitcher[i]} + u_{year[i]} + \epsilon_i$$

Positive interaction coefficients indicate that higher values of a given trait are associated with a larger increase in offensive production at Coors Field, while negative coefficients indicate a smaller or diminished effect.

As in the archetype model, random intercepts for batter and pitcher identity are included to account for differences in baseline ability, and year fixed effects capture variation in the overall run environment across seasons. The Rockies indicator is included to separate park effects from team-specific influences.

This specification provides a more flexible and detailed framework for understanding Coors Field effects. By allowing the park effect to vary continuously with underlying skill traits, the model identifies which specific offensive characteristics are most strongly associated with changes in performance at altitude, rather than relying on coarse group-level differences.

3. Pitch Shape Interaction Model

To evaluate whether the Coors Field environment differentially affects specific pitch types, the analysis incorporates a pitch shape interaction model based on pitch-level clustering. Rather than relying on traditional pitch labels, this approach groups pitches according to their underlying physical characteristics, allowing for a more precise representation of pitch behavior.

Pitch shape clusters are constructed using pitch-level Statcast data, with velocity, spin rate, induced vertical break (IVB), and horizontal break (HB) used as the primary inputs. These variables are standardized prior to clustering to ensure comparability across features. To improve computational efficiency while maintaining representation across pitch types, a stratified sample of pitches is drawn by pitch bucket, and a GMM is estimated on this sample. The number of clusters is selected by comparing model fit across candidate specifications.

The fitted GMM is then used to assign a pitch shape cluster to every pitch in the full dataset. Each cluster represents a distinct pitch shape profile characterized by a combination of velocity, spin, and movement properties. Cluster centers are examined to ensure interpretability and to associate clusters with known pitch types (e.g., four-seam fastballs, sinkers, sliders, or sweepers).

The pitch shape interaction model is estimated at the plate appearance level using run value as the response variable. The specification includes pitch usage proportions and their interactions with the Coors Field indicator:

$$RunValue_i = \beta_0 + \beta_1 * Coors_i + \sum_k \beta_k * Pitch_{k,i} + \sum_k \delta k * (Coors_i * Pitch_{k,i}) + \beta_4 * Rockies_i + u_{batter[i]} + u_{pitcher[i]} + u_{year[i]} + \epsilon_i$$

where $Pitch_{k,i}$ represents the proportion of pitches from cluster k used by the pitcher. Random intercepts for pitcher and batter identity account for differences in baseline ability, and year fixed effects capture variation in the overall run environment. This framework allows the analysis to test whether pitchers who rely more heavily on certain pitch shapes experience different outcomes at Coors Field.

4. Pitcher Archetype Interaction Model

While the pitch shape interaction model focuses on the effects of individual pitch types, the pitcher archetype model examines whether broader pitching styles respond differently to the Coors Field environment. This approach groups pitchers based on their overall pitch mix and repertoire composition rather than evaluating individual pitch characteristics in isolation.

Pitcher archetypes are constructed using pitcher-season level summaries of pitch shape usage. For each pitcher, the proportion of pitches thrown from each pitch shape cluster is calculated, producing a vector that describes the composition of the pitcher's arsenal. In addition to pitch usage, an entropy measure is computed to capture arsenal diversity. Higher entropy values indicate a more balanced pitch mix, while lower values indicate reliance on a smaller number of pitch types.

These features are combined to form a pitcher-level representation that captures both pitch selection and repertoire diversity. A GMM is then applied to these features to group pitchers into clusters representing distinct pitching archetypes. The number of clusters is selected based on model fit and interpretability, resulting in a set of archetypes that reflect differences in pitch mix and strategic approach.

Each plate appearance is linked to the archetype associated with the pitcher's player-season, allowing offensive outcomes to be analyzed as a function of pitching style. The impact of Coors Field across pitcher archetypes is then estimated using a mixed-effects regression model:

$$RunValue_i = \beta_0 + \beta_1 * Coors_i + \beta_2 * Cluster_i + \beta_3 * (Coors_i * Cluster_i) + \beta_4 * Rockies_i + u_{batter[i]} + u_{pitcher[i]} + u_{year[i]} + \epsilon_i$$

As in previous models, random intercepts for pitcher and batter identity control for baseline differences in performance, while year fixed effects account for changes in the league run environment over time. This specification provides a higher-level perspective on whether certain pitching strategies are more or less susceptible to the effects of altitude.

5. Pitch Physics Analysis

In addition to evaluating hitter and pitcher traits, the analysis examines how pitch characteristics change at Coors Field due to the stadium's high altitude. Reduced air density weakens aerodynamic forces, particularly the Magnus effect, which plays a central role in generating pitch movement. As a result, pitches thrown at altitude may exhibit altered vertical and horizontal movement, potentially affecting both pitch effectiveness and hitter outcomes.

Pitch-level Statcast data is used to construct a dataset of pitch characteristics, including velocity, spin rate, induced vertical break (IVB), and horizontal break (HB). These variables capture the physical behavior of pitches as they travel toward the plate. Observations with extreme or implausible values are removed, and horizontal movement is adjusted for pitcher handedness to ensure consistent interpretation of pitch direction.

To isolate the effect of the Coors environment, pitch characteristics are compared within pitcher-season and pitch-type combinations. For each pitcher and pitch type, average pitch characteristics are calculated separately for pitches thrown at Coors Field and those thrown in other ballparks. Only pitcher-seasons with sufficient observations in both environments are retained to ensure reliable comparisons.

Differences between Coors and non-Coors environments are then computed at the pitcher-season level, measuring how pitch behavior changes when the same pitcher throws the same pitch type under different conditions. These within-pitcher differences are subsequently aggregated across pitchers to produce league-level estimates of how pitch characteristics shift at Coors Field for each pitch type.

This approach provides a direct measurement of how pitch movement and velocity change at altitude while controlling for pitcher-specific tendencies. By isolating environment-driven differences in pitch behavior, the analysis offers a physical explanation for observed changes in offensive outcomes at Coors Field and complements the trait-based and clustering-based models used elsewhere in the study.

5. Results

1. Hitter Archetype Interaction Model

Clustering based on Statcast-derived traits identifies three distinct hitter archetypes that differ in contact quality, swing tendencies, and overall offensive approach. These clusters reflect meaningful variation in how hitters generate offense, particularly along the dimensions of power, contact ability, and swing aggressiveness.

Hitter Archetype Clusters

Cluster	Avg EV	90th Percentile EV	Pull%	Air Pull%	Barrel%	Zone Swing%	Zone Contact%	K%	BB%	Sprint Speed
1	87.91602	104.2341	20.30%	14.64%	7.97%	67.18%	83.51%	22.66%	7.52%	27.42728
2	89.68590	104.9248	20.65%	15.00%	11.36%	65.95%	82.76%	23.75%	10.23%	26.67418
3	87.99956	103.1789	15.50%	9.28%	5.64%	64.74%	87.48%	18.31%	7.52%	27.18177

Cluster 1 is characterized by an aggressive, pull-oriented profile. These hitters exhibit relatively high zone swing rates (67.18%) and above-average pull tendencies, particularly on airborne contact. While their average exit velocity is comparable to other groups, they display moderate barrel rates and lower contact rates, indicating a more aggressive approach that prioritizes driving and pulling the ball when contact is made.

Cluster 2 represents a power-oriented profile with the strongest contact quality metrics. This group records the highest average exit velocity (89.7 mph), highest 90th percentile exit velocity (104.9 mph), and highest barrel rate (11.4%) among the clusters. These hitters also maintain relatively aggressive swing behavior, though with slightly lower zone contact rates and higher strikeout rates, consistent with a power-focused offensive approach.

Cluster 3 reflects a contact-oriented profile. These hitters exhibit the highest zone contact rate (87.5%) and the lowest strikeout rate (18.3%), along with lower barrel rates (5.6%) and weaker top-end exit velocity. This group also shows lower pull tendencies and a more all-fields approach, supported by high sprint speed. Overall, this profile emphasizes contact ability and bat control over power.

To assess whether these archetypes experience different outcomes at Coors Field, a mixed-effects regression model is estimated using plate appearance-level run value as the response variable. The model includes interaction terms between the Coors Field indicator and hitter cluster, allowing the estimated park effect to vary across archetypes.

Variable	Coefficient	Std. Error	t-value
Coors × Cluster 1	0.0482	0.0042	11.36
Coors × Cluster 2	-0.0219	0.0060	-3.65
Coors × Cluster 3	-0.0157	0.0053	-2.96

Mixed Effects Model: Coors Field × Hitter Archetype

Consistent with the well-documented offensive environment at Coors Field, the baseline Coors effect is positive and highly significant, indicating a substantial increase in run value per plate appearance. However, the interaction results show that this effect is not uniform across hitter types.

Cluster 1 serves as the reference group, with an estimated Coors effect of approximately +0.048 runs per plate appearance ($t = 11.36$). Relative to this group, both Cluster 2 and Cluster 3 exhibit negative interaction coefficients (−0.0219 and −0.0157, respectively), indicating that their gains at Coors Field are smaller than those of Cluster 1 hitters. These differences are statistically significant, suggesting that the magnitude of the Coors effect does vary across archetypes.

However, while these differences are statistically detectable, they are modest in magnitude. All three clusters still experience positive Coors effects, and the variation across archetypes is relatively small compared to the overall increase in offensive production at Coors Field. This suggests that while hitter style influences the degree of benefit, the altitude-driven environment broadly increases offense across all hitter types.

Model Comparison: Baseline vs Cluster Interaction Model

Likelihood Ratio Test

Model	**Parameters**	**AIC**	**BIC**	**Log Likelihood**	**χ²**	**df**	**p-value**
Baseline Model	NA	1,829,075	1,829,295	–914,519	NA	NA	NA
Cluster Interaction Model	2	1,829,064	1,829,308	–914,512	15	2	5.27e-04

Model comparison results reinforce this interpretation. A likelihood ratio test comparing the cluster interaction model to a baseline specification without interaction terms yields a statistically significant improvement in fit ($\chi^2 \approx 15$, $p < 0.001$). This indicates that allowing the Coors effect to vary by archetype provides additional explanatory power, though the overall contribution remains limited.

Taken together, these results suggest that hitter archetypes capture meaningful differences in offensive style, and these differences translate into modest variation in Coors Field performance. However, the relatively small magnitude of these effects indicates that discrete archetypes alone do not fully explain variation in offensive outcomes at Coors Field, motivating the use of continuous trait-based models in the next section.

2. Hitter Trait Interaction Model

While the clustering analysis provides an intuitive way to summarize hitter archetypes, these clusters represent a relatively coarse summary of offensive skill. To more precisely evaluate how individual batted-ball characteristics interact with the Coors Field environment, a mixed-effects regression model is estimated using continuous hitter traits rather than discrete cluster labels.

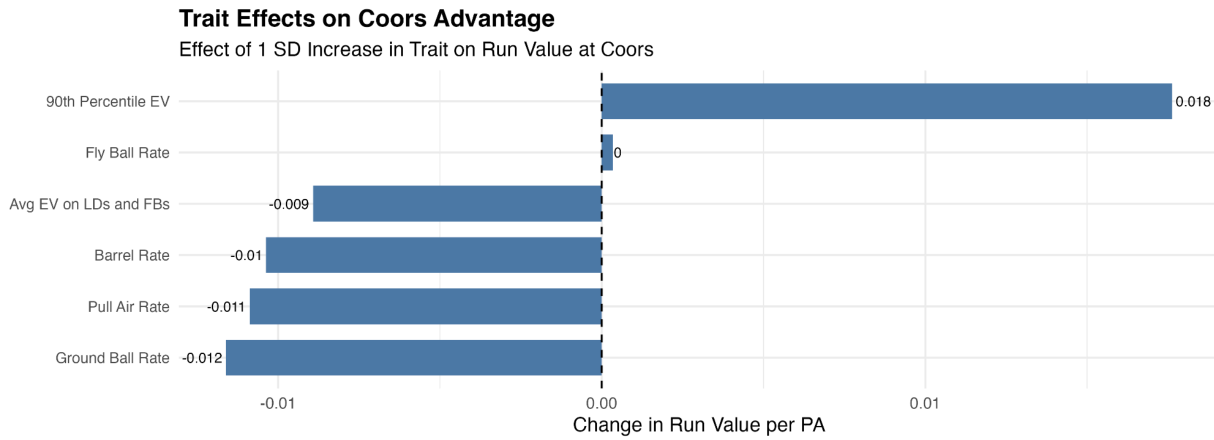
The model includes interactions between the Coors Field indicator and several key offensive traits, including 90th percentile exit velocity, ground ball rate, fly ball rate, average exit velocity on line drives and fly balls, barrel rate, and pull air rate. As in the archetype model, random intercepts for batter and pitcher identity are included to account for baseline differences in player ability, along with year effects to capture variation in the run environment across seasons.

Model Comparison

Trait Model vs Baseline

Model	AIC	χ^2	p-value
Baseline	1,829,301	NA	NA
Trait Model	1,828,067	1,239	< 2.2e-16

Comparing the continuous trait model to the baseline specification shows a substantial improvement in model fit. The likelihood ratio test strongly favors the trait interaction model ($\chi^2 \approx 1239$, $p < 2.2e\text{-}16$), and the model reduces AIC by more than 1,200 points. This indicates that continuous hitter traits explain a significant portion of the variation in offensive outcomes at Coors Field beyond a simple park effect.



The estimated interaction coefficients identify which offensive traits are most strongly associated with changes in performance at Coors Field. Among the traits examined, 90th percentile exit velocity exhibits the largest positive interaction effect, indicating that hitters capable of generating elite top-end contact quality experience the greatest increases in run value at altitude. This result is consistent with the underlying physics of reduced air density, where well-struck balls travel farther and are more likely to result in extra-base hits.

In contrast, several traits display negative interaction coefficients, including ground ball rate, barrel rate, average exit velocity on airborne contact, and pull air rate. These coefficients indicate that hitters who rely more heavily on these characteristics experience smaller increases in run value at Coors Field relative to the baseline effect.

Importantly, these negative interaction effects do not imply that these hitters perform worse at Coors Field in absolute terms. The baseline Coors effect remains positive and substantial (approximately +0.035 runs per plate appearance for an average hitter), meaning that all hitter types still benefit from the Coors environment. Instead, the interaction terms reflect differences in the magnitude of that benefit. For example, a higher ground ball rate reduces the size of the Coors boost relative to a fly ball-oriented profile, rather than leading to an actual decline in offensive production.

Overall, the continuous trait model provides a more detailed and precise view of how offensive skill interacts with the Coors Field environment. Rather than relying on broad archetype groupings, the model identifies specific traits that drive variation in performance, with 90th percentile exit velocity emerging as the most important factor. These results suggest that the Coors Field effect is better explained by continuous skill characteristics than by discrete hitter archetypes.

3. Pitch Shape Interaction Model

To evaluate whether the Coors Field environment differentially affects specific pitch types, pitch-level data is first grouped into clusters based on underlying physical characteristics. Using velocity, spin rate, induced vertical break (IVB), and horizontal break (HB), a Gaussian Mixture

Model identifies seven distinct pitch shape clusters that capture variation in pitch movement and velocity profiles.

Pitch Shape Clusters

Cluster Centers Based on Velocity, Spin, and Movement

Cluster	Velocity (mph)	Spin Rate (rpm)	IVB (in)	HB (in)	Count
1	87.6	2,364.0	6.2	−1.6	1,607,048
2	94.0	2,281.0	17.6	9.0	2,215,067
3	85.5	1,713.0	6.9	15.2	964,585
4	81.8	2,562.0	0.7	−13.1	433,499
5	84.0	1,775.0	4.5	5.6	141,750
6	79.1	2,535.0	−11.4	−8.3	612,248
7	93.4	2,164.0	10.2	16.9	1,233,697

Because the clustering procedure is entirely data-driven, the resulting clusters are not labeled. To interpret the clusters, domain knowledge of pitch shapes is applied to the cluster centers. Based on their average velocity, spin, and movement characteristics, the clusters are identified as follows (Clusters 1 through 7): Slider/Cutter, 4-Seam Fastball, Changeup, Sweeper, Splitter, Curveball, and Sinker. This mapping allows the clusters to be linked to recognizable pitch types while preserving the underlying data-driven structure.

The table above shows that these clusters capture distinct pitch profiles. For example, the 4-seam fastball cluster exhibits the highest velocity and induced vertical break, while the curveball cluster is characterized by large negative vertical and horizontal movement.

To examine how these pitch shapes interact with the Coors Field environment, a mixed-effects regression model is estimated using pitch shape usage at the plate appearance level. The model includes interaction terms between the Coors Field indicator and the proportion of each pitch shape in a pitcher’s repertoire, allowing the estimated Coors effect to vary based on pitch selection.

Pitch Type Effects at Coors Field

Mixed Effects Model: Run Value per PA

Variable	Coefficient	Std. Error	t-value
Coors × Cluster 1	0.022	0.009	2.52
Coors × Cluster 2	0.026	0.015	1.79
Coors × Cluster 3	-0.003	0.017	-0.17
Coors × Cluster 4	0.011	0.021	0.52
Coors × Cluster 5	-0.008	0.035	-0.22
Coors × Cluster 6	0.028	0.024	1.20
Coors × Cluster 7	0.029	0.015	1.91

The interaction results indicate that differences across pitch shapes are generally limited. Most coefficients are small in magnitude and not statistically significant, suggesting that the Coors Field effect does not vary substantially based on pitch shape usage.

One exception is Cluster 1 (slider/cutter), which shows a statistically significant positive interaction effect (coefficient ≈ 0.022, t ≈ 2.5). This indicates that pitchers who rely more heavily on slider/cutter-type pitches allow slightly more runs at Coors Field than the baseline Coors effect would predict. However, the magnitude of this effect remains modest.

Other pitch shapes, including four-seam fastballs, sinkers, and curveballs, show small positive interaction coefficients, while changeups and splitters exhibit slightly negative coefficients. However, these estimates are not statistically distinguishable from zero, indicating that these differences are not reliably different from the baseline Coors effect.

Overall, these results suggest that the impact of Coors Field is broadly consistent across pitch shapes. While there is some minor variation in how different pitch types perform at altitude, these differences are small relative to the overall increase in offensive production. This contrasts with the hitter-side trait model, where specific offensive characteristics play a much larger role in explaining variation in Coors performance.

4. Pitcher Archetype Interaction Model

While the pitch shape interaction model evaluates the impact of individual pitch types, it does not account for how pitches are combined within a pitcher's overall arsenal. To address this, pitchers are grouped into archetypes based on their pitch mix, defined by the proportion of each pitch shape used and overall arsenal diversity.

Pitcher Archetype Clusters Average Pitch Mix (%) and Arsenal Diversity									
Cluster	**4-Seam**	**Sinker**	**Slider/Cutter**	**Curveball**	**Sweeper**	**Changeup**	**Splitter**	**Entropy**	**# Pitchers**
1	49%	3%	22%	11%	0%	13%	1%	1.20	631
2	28%	18%	18%	12%	11%	12%	0%	1.48	1,006
3	21%	14%	16%	6%	12%	20%	11%	1.36	516
4	35%	16%	22%	7%	1%	15%	3%	1.46	805
5	24%	26%	24%	1%	20%	5%	0%	1.18	528
6	34%	24%	27%	0%	0%	15%	1%	1.07	452
7	8%	40%	20%	10%	1%	20%	1%	1.28	447
8	42%	0%	39%	14%	2%	2%	0%	0.96	435

Pitcher archetypes are constructed using a GMM applied to pitch shape usage rates and entropy, which captures how diversified a pitcher's repertoire is. The resulting clusters represent distinct pitching styles, ranging from fastball-heavy profiles to more balanced arsenals with a wider distribution of pitch types.

The cluster summary highlights clear differences in pitch mix composition across groups. For example, some clusters are dominated by four-seam fastballs, while others rely more heavily on sinkers, slider/cutter combinations, or changeup usage. Differences in entropy further distinguish pitchers with highly specialized arsenals from those with more balanced pitch distributions.

To evaluate whether these pitching styles respond differently to the Coors Field environment, a mixed-effects regression model is estimated with interaction terms between the Coors Field indicator and pitcher archetype.

Pitcher Archetype Effects at Coors Field Mixed Effects Model: Run Value per PA			
Variable	**Coefficient**	**Std. Error**	**t-value**
Coors × Cluster 1	0.032	0.005	6.46
Coors × Cluster 2	0.012	0.007	1.86
Coors × Cluster 3	0.002	0.008	0.19
Coors × Cluster 4	0.004	0.007	0.56
Coors × Cluster 5	0.000	0.008	0.00
Coors × Cluster 6	0.009	0.008	1.07
Coors × Cluster 7	0.005	0.008	0.63
Coors × Cluster 8	0.008	0.008	0.99

The results show that Coors Field increases run value across all pitcher archetypes, consistent with the overall offensive environment observed throughout the analysis. The baseline Coors effect remains positive, indicating that all pitchers experience some degree of reduced run prevention at altitude.

Across archetypes, however, there is limited evidence of meaningful differences in the magnitude of this effect. Most interaction coefficients are small and not statistically significant, suggesting that the Coors effect is broadly similar across different pitching styles.

One exception is Cluster 1, which shows a statistically significant positive interaction (coefficient ≈ 0.032, $t \approx 6.46$). This indicates that pitchers in this group allow slightly more runs at Coors Field relative to the baseline. Based on the cluster summary, this group is characterized by a heavy reliance on four-seam fastballs and slider/cutter combinations, limited use of curveballs and changeups, along with relatively low pitch diversity.

Other clusters show smaller positive interaction effects, but these are not statistically distinguishable from zero. This suggests that while pitching styles differ in composition, these differences do not meaningfully alter how pitchers are affected by the Coors environment.

Overall, the pitcher archetype results reinforce the findings from the pitch shape interaction model. While there is minor variation across pitching styles, the effect of Coors Field is largely consistent across both pitch types and pitcher archetypes. This indicates that altitude-driven changes in pitch behavior impact pitchers broadly, rather than disproportionately affecting specific arsenals.

5. Pitch Physics Analysis

While the hitter and pitcher models identify which players benefit more or less from the Coors Field environment, those differences ultimately stem from how pitches behave at altitude. Reduced air density weakens aerodynamic forces acting on the baseball, particularly the Magnus effect, leading to systematic changes in pitch movement.

To quantify these changes, pitch tracking data is used to compare pitch characteristics between Coors Field and non-Coors environments within pitcher-season and pitch-type combinations. This within-pitcher comparison isolates how the same pitcher’s pitches behave differently at altitude, controlling for differences in individual pitching style.

Pitch Changes at Coors Field Average Difference Between Coors and Non-Coors Environments				
Pitch Type	Δ Velocity (mph)	Δ Spin (rpm)	Δ IVB (inches)	Δ HB (inches)
Four-Seam	-0.03	9.70	-3.21	2.00
Curveball	-0.03	20.40	1.95	-1.75
Cutter	0.01	7.21	-1.82	0.28
Sweeper	-0.07	7.27	-1.60	-2.61
Sinker	-0.02	12.01	-1.50	3.59
Changeup	-0.06	17.63	-1.20	3.59
Slider	0.05	27.46	-0.59	-0.60
Splitter	-0.10	5.02	-0.26	3.40

Across pitch types, the most consistent effect is a reduction in induced vertical break (IVB). Four-seam fastballs show the largest decrease, with IVB dropping by approximately 3.2 inches at Coors Field, reflecting the loss of vertical "ride," and this loss of IVB is reflected across all other pitch types as well, other than curveballs, which experience a gain of IVB. However, curveballs rely on negative IVB, so a gain in IVB actually means the curveball is losing break and becoming less sharp.

Changes in horizontal movement are more heterogeneous. Curveballs, sweepers, and sliders tend to lose horizontal break, while sinkers and changeups gain arm-side movement. These differences indicate that altitude does not simply reduce movement uniformly, but instead alters the shape of pitches in ways that depend on their underlying spin and movement profiles.

Velocity and spin rate show minimal and inconsistent changes across pitch types, indicating that the primary environmental effect operates through movement rather than pitch speed or spin generation. Taken together, these results suggest that Coors Field systematically alters pitch movement, with particularly meaningful reductions in vertical break.

To understand how these physical changes relate to performance, pitch-level regression models are used to estimate how pitch characteristics are associated with run value at Coors Field. These models are estimated separately by pitch type and include interactions between Coors Field and standardized pitch characteristics such as velocity, spin, IVB, and horizontal break. The estimated effects are then scaled to a seasonal context to aid interpretation.

Seasonal Run Impact of Pitch Traits at Coors Field

Runs Saved/Lost Over a Season From a 1 SD Change in Each Pitch Characteristic

Pitch Type	Velocity	Spin	IVB	HB	VAA	HAA	Extension	Arm Angle
Four-Seam	-9.1	-2.1	-14.9	-14.2	-0.8	-1.8	-1.0	10.5
Sinker	-2.9	-1.9	8.5	7.4	5.1	0.8	-0.2	-3.4
Cutter	-0.8	-4.6	9.8	2.7	-6.4	0.0	-0.2	-6.9
Slider	-4.7	-1.3	-2.7	-7.9	-3.1	0.6	1.1	0.1
Sweeper	-8.0	0.1	-12.0	-7.1	4.9	1.1	-3.3	0.5
Curveball	-7.6	-1.4	1.9	-3.0	5.2	0.5	-3.2	0.8
Changeup	-0.2	-0.8	12.7	8.5	2.1	0.2	-0.3	-5.4
Splitter	-10.6	4.4	17.4	5.7	-2.2	-6.5	0.1	-8.2

The table reports the estimated change in runs allowed over a full season associated with a one standard deviation increase in each pitch characteristic for a given pitch type at Coors Field. These values are derived from the regression model and scaled by typical pitch usage, and should therefore be interpreted as approximate, model-based sensitivities rather than direct causal effects.

Several patterns emerge from these estimates. Four-seam fastballs appear particularly sensitive to vertical movement, with increases in IVB associated with substantial improvements in run prevention. Because IVB is consistently reduced at Coors Field, this implies a loss of effectiveness for four-seam fastballs in the high-altitude environment. Sweeper-type pitches show a similar dependence on movement, with reductions in both vertical and horizontal break associated with worse outcomes.

In contrast, sinkers and changeups show positive associations with horizontal movement, suggesting that increases in arm-side run may improve run prevention for these pitch types. Since these pitches tend to gain horizontal movement at Coors Field, they may be less negatively impacted relative to pitches that rely heavily on vertical ride.

Importantly, these results should be interpreted in relative terms. They do not imply that any pitch type becomes ineffective at Coors Field, but rather that altitude-driven changes in movement alter the degree to which certain pitch characteristics contribute to run prevention.

Overall, the pitch physics analysis provides a physical explanation for the patterns observed in the earlier models. Reduced vertical movement, particularly for four-seam fastballs and sweeping breaking pitches, contributes to increased offensive production. At the same time, because these movement changes affect nearly all pitchers, the overall impact of Coors Field is broadly consistent across pitch types and pitcher archetypes. This helps explain why the pitch shape and pitcher archetype models show relatively limited differences, despite clear underlying changes in pitch behavior.

6. Discussion

The results of this study provide a more detailed understanding of how the Coors Field environment affects performance by linking player outcomes to both skill characteristics and underlying pitch physics. While it is well established that Coors Field increases offensive production, the analysis shows that this effect is not entirely uniform. There is meaningful variation in how players experience the Coors environment, but that variation is structured and can be explained by specific offensive traits and changes in pitch behavior rather than by broad player archetypes alone.

One of the central findings is the difference in explanatory power between discrete archetypes and continuous skill traits. The clustering analysis identifies intuitive hitter groups that differ in approach, contact quality, and swing behavior, and these groups do exhibit modest differences in Coors Field performance. However, the magnitude of these differences is relatively small, even when statistically significant. In contrast, the continuous trait interaction model provides a much stronger explanation of variation in offensive outcomes, as reflected in the large improvement in model fit. This suggests that while archetypes are useful for summarizing player types, they oversimplify the underlying variation in skill. Performance at Coors Field is better understood as a function of specific traits, particularly top-end exit velocity, rather than membership in a discrete archetype.

The importance of individual traits aligns closely with the physical changes observed in pitch behavior at altitude. The pitch physics analysis shows that induced vertical break is consistently reduced across pitch types at Coors Field, with particularly large decreases for four-seam fastballs. Because vertical ride is a key component of fastball effectiveness, this reduction makes it more difficult for pitchers to miss bats or generate weak contact. As a result, hitters who are capable of producing high-quality contact, especially those with strong top-end exit velocity, are better positioned to take advantage of these conditions. This provides a clear mechanism linking the statistical results from the hitter models to the underlying physics of the environment.

At the same time, the pitching-side results show relatively limited variation across pitch types and pitcher archetypes. The pitch shape interaction model finds that most pitch types experience similar changes in effectiveness at Coors Field, with only minor differences across clusters. A similar pattern emerges in the pitcher archetype model, where different pitch mixes and levels of arsenal diversity do not substantially alter the magnitude of the Coors effect. These findings suggest that altitude-driven changes in pitch movement affect nearly all pitchers in a similar way. Rather than disproportionately impacting specific pitch types or pitching styles, the Coors environment creates a broad shift in pitch behavior that leads to increased offense across the board.

This combination of results helps explain why hitter-side variation is more pronounced than pitcher-side variation. On the pitching side, the physical environment imposes similar constraints on all pitchers by reducing movement. On the hitter side, however, players differ in how well their skill sets allow them to capitalize on those changes. Hitters who rely on high-quality contact and pulled batted balls are better able to convert reduced pitch movement into increased offensive production, while hitters with less power or more ground-ball-oriented profiles see smaller gains. This asymmetry highlights the importance of considering both environmental effects and player-specific traits when evaluating performance.

From a practical perspective, these findings have implications for player evaluation and roster construction. For hitters, the results suggest that traits related to contact quality, particularly top-end exit velocity, are especially valuable in high-altitude environments. Teams may benefit from prioritizing hitters who can consistently generate hard contact when building lineups for Coors Field. For pitchers, the results indicate that no single pitch type or arsenal composition fully mitigates the effects of altitude. However, pitchers who rely less on vertical movement and more on horizontal movement, such as those who feature sinkers or changeups, may be somewhat less affected than those who depend heavily on four-seam fastballs or sweeping breaking pitches. More broadly, the findings suggest that park adjustments and player evaluations should account for how specific traits interact with environmental conditions rather than relying solely on average park effects.

Several limitations should be noted. First, Coors exposure is relatively limited for most players, which introduces noise in estimating player-specific Coors effects. Because most players only accumulate a small number of plate appearances or pitches at Coors each season, these estimates can be sensitive to short-term performance variation rather than true underlying skill, making it difficult to cleanly separate signal from noise at the individual level. Second, pitch clustering simplifies pitch shape into discrete groups, which may mask more nuanced differences in pitch behavior. While clustering helps make the analysis more interpretable, it forces a continuous range of pitch characteristics into a small number of categories, meaning that subtle but potentially important differences in movement profiles or pitch quality within a cluster are not fully captured.

Overall, this study demonstrates that the impact of Coors Field is best understood as the result of an interaction between environmental physics and player skill. While altitude increases offense across all players, the extent of that increase depends on how individual traits interact with changes in pitch movement. By combining Statcast-based skill measures with pitch-level physics, the analysis provides a more complete framework for understanding performance in extreme environments.

7. Conclusion

This study examines how the Coors Field environment interacts with player skill by combining Statcast-based trait analysis with pitch-level physics. While it is well established that Coors Field increases offensive production, the results show that this effect is not entirely uniform. There is meaningful variation in how players experience the Coors environment, but that variation is better explained by continuous skill traits and changes in pitch behavior than by broad player archetypes.

On the hitter side, discrete archetypes capture intuitive differences in offensive approach, but they provide only a limited explanation of Coors Field performance. Continuous trait models offer a more precise view, showing that specific skills, particularly top-end exit velocity, play a central role in determining which hitters benefit most from altitude. This highlights the importance of evaluating players based on the underlying characteristics that drive performance rather than relying solely on categorical groupings.

On the pitching side, both pitch shape and pitcher archetype analyses reveal relatively consistent Coors effects across different pitch types and arsenal constructions. These findings suggest that altitude-driven changes in pitch movement affect most pitchers in a similar way, limiting the extent to which pitch selection or overall repertoire can mitigate the Coors environment.

The pitch physics analysis provides a unifying explanation for these results. Reduced air density leads to systematic changes in pitch movement, particularly a loss of vertical break, which contributes to increased offensive production. Because these movement changes occur broadly across pitch types, they create a general increase in offense rather than large differences across pitching styles. At the same time, hitters vary in their ability to take advantage of these conditions, leading to the trait-driven differences observed in the offensive models.

Overall, the findings show that Coors Field effects are driven more by the interaction between environmental physics and player skill than by player archetype alone. This framework offers a more complete understanding of performance in extreme environments and suggests that player evaluation and park adjustments should account for how specific traits interact with environmental conditions.